

Custody in Digital Finance

Turnkey’s practical guide to custody and non-custody for digital assets

turnkey

Turnkey delivers secure wallet infrastructure, powered by battle-tested Trusted Execution Environments (TEEs), designed for developers, and ready to scale with your applications. Equip your web3 customers with trusted wallet security that is isolated, verifiable, and tamper-resistant.

This ebook is for informational and educational purposes only and does not constitute legal, financial, or regulatory advice. "Non-custodial" as used throughout describes Turnkey's architectural design — specifically, that neither Turnkey nor customer application infrastructure holds raw private key material or can unilaterally authorize transactions. This is a technical characterization, not a regulatory conclusion. Turnkey makes no representation regarding the regulatory or custody classification of any application built on its infrastructure. Custody classification and licensing requirements vary by jurisdiction, regulatory framework, and product configuration. Application builders should consult qualified legal counsel to determine their specific custody and regulatory obligations.

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What is crypto custody?

Custody in traditional finance usually begins with an institution.

In that model, banks, brokers, and trust companies manage money movement from the start of a transaction until the funds settle in a user's account. Customers may own the asset on paper, but in traditional finance, the institution controls how assets move through the financial system.

Crypto custody works differently. In traditional finance, custody means a regulated institution holds assets on your behalf and a chain of institutional records determines who owns what. Control in crypto is based on cryptographic authorization instead. A blockchain does not check in with a bank before moving assets. It verifies whether a transaction was signed with the correct credentials. These credentials provide cryptographic authorization to approve onchain money flows.

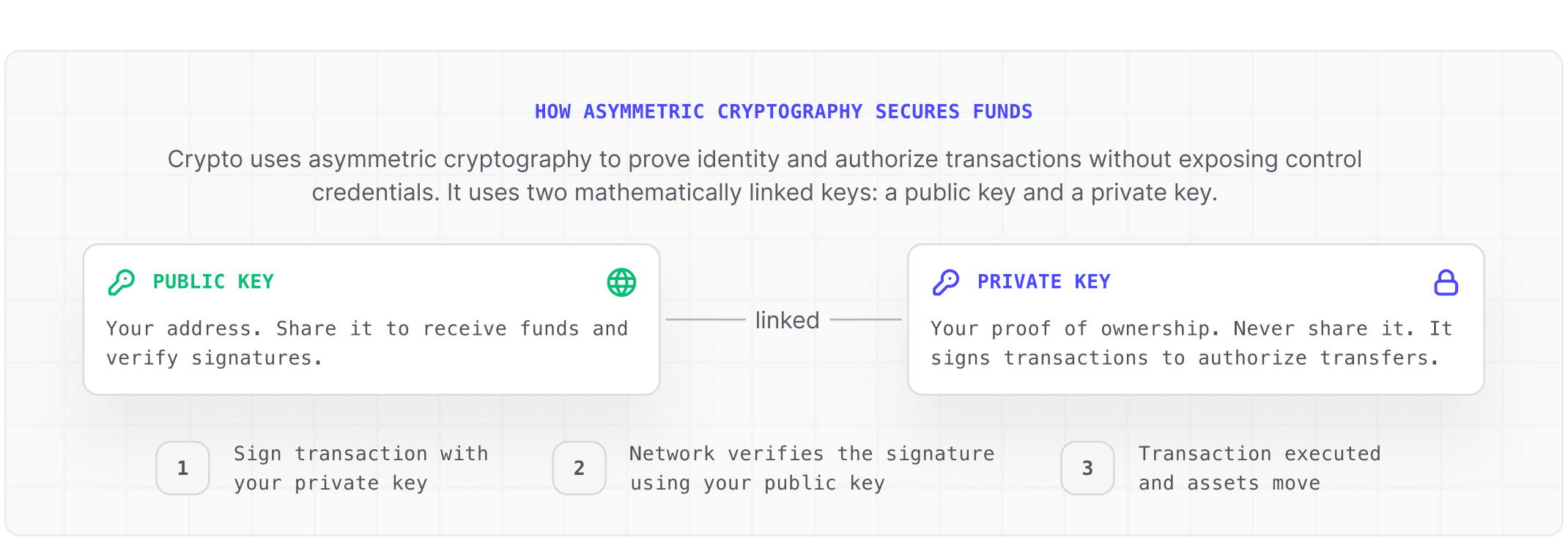
In crypto, we call these credentials private keys. Private keys are long alphanumeric values that prove control over assets and authorize their use onchain.

Any party that produces a valid signature with a private key has practical control over the asset, regardless of what any legal agreement says. If the signature is valid and the network accepts the transaction, the asset moves.

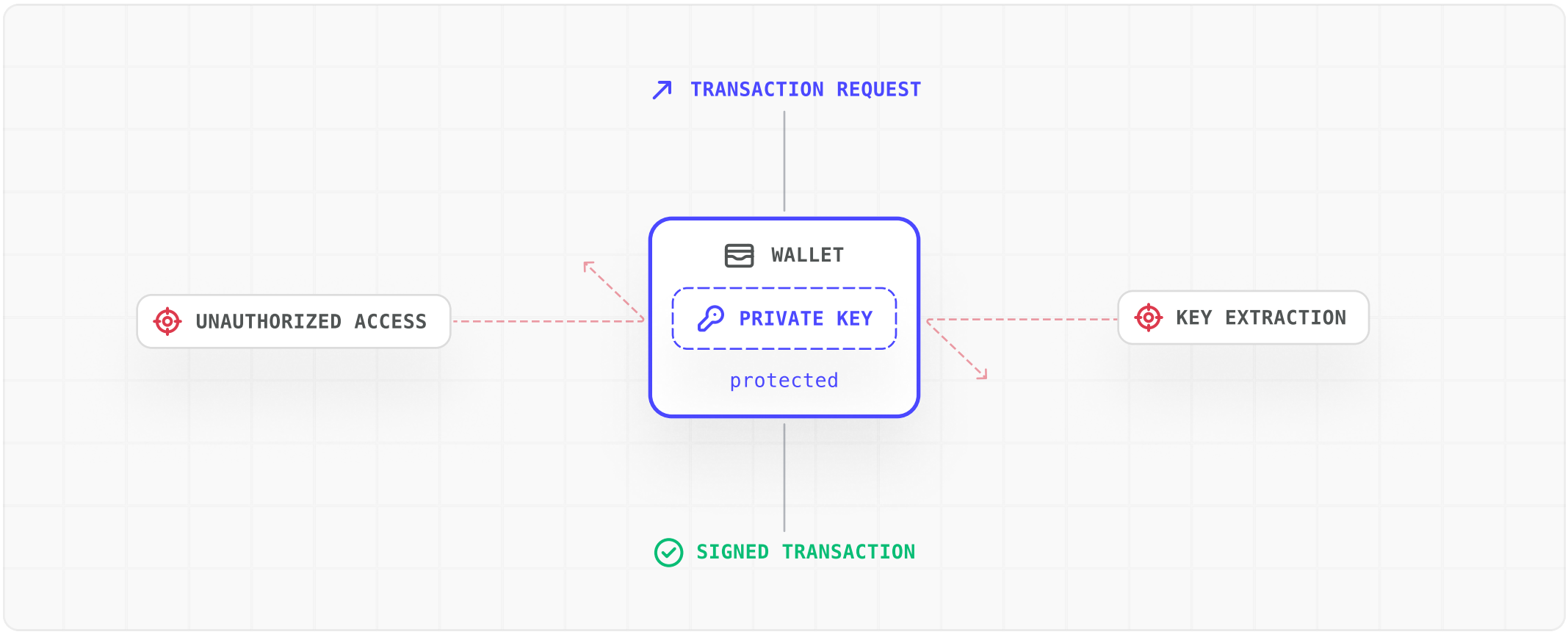
This is why traditional custody assumptions do not transfer cleanly to crypto. In conventional markets, errors can often be reviewed, reversed, or corrected through institutional processes that sit between the customer and the asset.

In crypto, settlement can happen in seconds across global networks and often there is no central party that can intervene. A compromised private key or unauthorized signature becomes final before anyone has time to respond. Largely, then, crypto custody is an architectural problem. Much depends on how private keys are generated and protected, who can access them, and what controls are enforced before assets move.

For application builders, it is fundamental to understand who controls their users' private keys.



Wallets in crypto custody



In crypto, a wallet's primary role is to manage the custody layer for digital assets, and that starts with protecting private keys. Across crypto custody models, the core issue is signing authority: who can authorize a transaction, where does that signing authority live, and how easily can it be compromised.

Wallets can feel easy to use while hiding weak custody assumptions underneath. A product can give users a smooth login flow while placing too much practical control in the hands of the application, the provider, or a backend system.

Strong custody models make controls clear, limit key exposure, and enforce authorization rules as close to signing as possible. This is one of the reasons wallet infrastructure is such an important decision for application teams. In addition to shaping the user experience, it defines who has practical control over assets, how transactions are approved, what happens during recovery, and how much damage a compromised system can cause.

The right model should make wallets easy to use without weakening custody. It should let applications support onboarding, recovery, permissions, and policy controls while keeping signing authority in the right hands.

Three wallet custody models for modern financial applications

Applications and institutions typically choose from three custody models when scoping out their wallet infrastructure. These models include:

Custodial

A third party controls the keys

A third party controls the private keys or has practical authority over asset movement. The user may still own title to the assets legally, but the provider operates the signing environment and can authorize transactions on the user's behalf. This can make onboarding, recovery, and support easier, but it also concentrates trust in the provider.

PLATFORM WALLET

Risk: server compromise, insider access, or operational failure can expose keys

Non-custodial

user controls the keys

The user, institution, or end customer retains signing authority, while infrastructure handles secure key generation, storage, policy enforcement, and signing. This creates a middle path for teams that want stronger control without forcing every customer or internal team to manage the full signing stack manually.

USER WALLET

Risk: user trusts infrastructure security without holding custodial control

Hybrid

Both control the keys by policy

Control is shared across parties, systems, or workflows depending on the product, regulatory model, or operational requirements. A business might use non-custodial wallets for customer funds, custodial controls for treasury operations, and delegated signing rules for agents or internal teams.

PLATFORM USER

Risk: policy misconfiguration can blur the control boundary

A strong custody model makes the control boundary clear. It defines who can sign, what they are allowed to sign, where the key can exist, and what happens when access needs to be recovered or revoked. For builders, how signing authority is designed is the real custody question.